MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **Calculation**
1800 Security System
Mining Boundary Overall Perimeter Fence

Document Number:	**MD-216-1800-EG-CV-CAL-1001**
Revision:	**D01**
Revision Date:	**11-10-2023**
Status:	**AS BUILT**

Outotec Document Number:	900707	GAU01	ZZ01	CED01	00001	OU100660774	3
L&T Document Number:	O19045	900	C	K	24339	001	3

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	25.09.2019	FIRST ISSUE	SOR	SOR	SJM	
A02	08.11.2019	REVISED AS PER COMMENTS FROM EMPLOYER / PMC DATED 21.10.2019 AND AS MARKED.	SOR	SJM	AG	
CO1	10.02.2020	ISSUED FOR CONSTRUCTION	SOR	SJM	AG	
DO1	11.10.2023	AS BUILT	SOR	SJM	AG	

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (EPC-OT) and Larsen & Toubro (EPC-L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document

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1. INTENT

The document is prepared for the design calculation of the posts and foundation system for the overall mining boundary perimeter fence. The design calculation ensures the safety and serviceability of the fencing as well as the foundation system.

2. SCOPE

The design document is prepared for the overall mining boundary perimeter fence along with the foundation system. The document is prepared based on the following input documents :

a) MD-216-0000-EG-GE-DPP-0003 (Rev C04)_ Facility Boundary Fence.
b) MD-216-1800-EG-ME-BOD-1001 (Rev A01)_Basis of design, Gold processing Plant security system

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3. DESIGN OF FENCE POST & FOUNDATION

Velocity Pressure , qz evaluated at height z shall be calculated by the following equation:

$q_z = 0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ kN/m^2 (Refer Cl. No 7.2.8 of SBC-301)

Basic Wind Speed	V =	177 km/h	
Velocity pressure exposure coefficient	Kz =	1.04	As per Table 7.2-2(Page-7/44)- Z=0 to 5m of SBC-301
Exposure Categories.	=	D	As per Project Basis of Design
Topographic Factor	K_{zt}=	1	As per Cl. 6.4.3.2

(As per Pt no 4 of cl 6.4.3.1 of SBC-301 , $H/L_h < 0.2$, hence topographic effect not considered)

Wind Directionality Factor	K_d =	0.85	As per Table 6.4-1(Page-6/7)

As the structure represent a low hazard to human life in the event of failure, it can be classified as category - I.

Importance factor	I =	0.77	As per Table 6.5-1(Page-6/9)
Hence,	qz =	1.01	kN/m^2

3.1 CHECK FOR BENDING STRESS OF FENCE POSTS

Force co-efficient for circular section (For the posts) =	C_f =	1.2	(As per Fig. 7.2-15 (Page 7/39)
For 20% solidity , Force Coefficient (for the wire mesh)	C_f =	1.3	(As per Fig. 7.2-17 (Page 7/41) of SBC-301

Total Load on vertical post due to wind considering projected frontal area of the post (Outer Dia = 60.3mm) ;

1.2x0.06x1.0x 1.01 + 1.3x0.2x(1x3) x1.01 =		0.86	kN/m =	58.9 lb/ft
Height of fence	=	1.55	m =	5.1 ft
Moment at top of footing ; M = 1.6 x {(0.86x1.55x1.55) /2} =		1.65	KN-m =	1651.75 N-m

(Considering load factor = 1.6 to be on the conservative side)

Shear at top of footing ; Q = (0.86 x 1.55) / 10 = 0.13 (Ton)

Section Properties of Hollow tube of OD 60.3mm (ASTM A-53 SCH.40) ;

Outer Diameter	d2 =	60.30	mm
Thickness	=	3.91	mm
Inner Diameter	d1 =	52.48	mm
Section Modulus, = 1/6 ($d2^3$-$d1^3$) =	Z =	12453.07	mm^3
Grade of Steel ; Fy	=	240.00	N/mm^2
Flexural design strength of beam = 0.9 x F_y x Z	=	2689.86	N-m (Cl. No 6.1.1
	>	1651.75	N-m of SBC-306:2007)

Hence, OK

3.2 CHECK FOR SHEAR STRESS OF FENCE POSTS

Design shear strength of post = 0.6 Ø F_n (Section 8.2 (b) of SBC-306:2007)

Ø = 0.90

Fn = Fy = 240 N/mm2

Design shear strength = 0.6x0.9x240 = 129.6 N/mm2

Shear area of the post section = [Π/4 ($d2^2$-$d1^2$)] / 2 = 346.16 mm2

Factored shear force = 1.6 x 0.13 = 0.2131 Ton = 2131.3 N

Shear stress = 2131.3 / 692.32 = 6.16 N/mm2 < 129.60 N/mm2 HENCE O.K.

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3.3 CHECK FOR COMPRESSION OF FENCE POSTS

Considering weight of pipe & chainlink for a span of 3m = 0.05 Ton
= 500 N

Considering load factor = 1.4 ; Factored design load = 1.4 x 500 = 700 N
(Load factor 1.4 is adopted as per clause 2.3.2 of SBC_301_2007)

Section Properties of Hollow tube of OD 60.3mm (ASTM A-53 SCH.40) ;

Outer Diamter	d2 =	60.30	mm
Thickness	t =	3.91	mm
Inner Diameter	d1 =	52.48	mm
Grade of Steel ; Fy	=	240	N/mm^2
Modulus of elasticity, E	=	200000	N/mm^2
Width Thickness ratio ; D/t	=	15.42	
Limiting width thickness ratio for compact section, $Y_p = 0.07 E/Fy$	=	58.33	(Table 2.5_1:SBC-306))
	>	15.42	

Flexural design strength of post = $\emptyset c P_n$ (Cl. No 5.2.1 of SBC-306:2007)

$\emptyset c$ = 0.85

P_n = $A_g F_{cr}$

A_g = gross area of member = $\Pi/4 (d2^2-d1^2)$ = 692.32 mm^2

Moment of Inertia, I = $\Pi /64(d2^4-d1^4)$ = 276506.7 mm^4

Radius of gyration about the axis of buckling , r = $\sqrt{I / A}$ = 19.98 mm

Effective length factor , K = 2

Laterally unbraced length of member, l = 1550 mm

$\lambda_c = (Kl / r\Pi) \sqrt{Fy / E}$ = 1.71 > 1.5

$F_{cr} = (0.877 / \lambda^2_c)) Fy$ = 71.87

Design strength of post = $\emptyset c P_n$ = 0.85x692.32x71.87 = 42295 N
> 700 N Hence,OK

3.4 CHECK FOR COMBINED STRESS OF FENCE POSTS

P_u = Required compressive strength = 500 N (Refer Section 8.1 of SBC-306:2007)

$\emptyset$ = Resistance factor for compression = 0.85

P_n = Nominal compressive strength = 42295 N

$P_u / \emptyset P_n$ = 0.014 < 0.2

$\emptyset_b$ = Resistance factor for flexure = 0.90

M_n = Nominal flexural strength determined in accordance with Section 6.1 of SBC-306 : 2007 = 2689862.9 N-mm

Mu = Required flexural strength determined in accordance with Section 3.1 of SBC-306 : 2007 = $B1M_{nt} + B2M_{lt}$
(As per cl 3.1.1 of SBC-306:2007)

M_{nt}=Required flexural strength in member assuming there is no lateral translation of the frame = 1032342 N-mm

B1 = $Cm / (1 - P_u/P_{e1})$ ⩾ 1 (As per cl 3.1.2 of SBC-306:2007)

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$P_{e1} = \Pi^2 EI / (KL)^2 =$ 56795.26 N

K = 2
L = 1550 mm
Cm = 0.85

B1 = 0.86 < 1
Hence , value to be adopted for ; B1 = 1

B2 = $1 / (1 - P_u/P_{e2})$
$P_{e2} = \Pi^2 EI / (KL)^2 =$ 56795.26 N

B2 = 1.01

Lateral deflection at the top of post ; s = $wl^4 / 8EI$ (w = uniformly distributed load on post
= 11.21 mm l = length of post)

M_{lt}=Required flexural strength in member as a result of lateral translation of the frame only = P_u x s
= 5606.1 N-mm

$Mu = B1M_{nt} + B2M_{lt} =$ 1037998 N-mm

Considering load combination of 1.2 DL + 1.6 WL as per SBC - 301

$(P_u / 2\emptyset P_n) + (M_u / \emptyset_b M_n) =$ 0.69 < 1 HENCE O.K
(As the lateral force acts only across the fence, there is no bending in fence post along the fence)

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3.5 STRESS CHECK FOR BACKSTAY AT STRAINER POSTS

Influence legth	=	90.00	m	
Height of fence	=	1.55	m	
Considering 20% solidity total wind force , P = 0.2x90x1.55x1.3x1.01x1000 =		36585	N	
Wind force along the axial direction of backstay = 36585 / Cos 45 ° =		51738	N	
Section Properties of Hollow tube of OD 42.2mm (ASTM A-53 SCH.40) ;				
Outer Diamter	D =	42.20	mm	
Thickness	t =	3.56	mm	
Inner Diameter	D1 =	35.08	mm	
Grade of Steel ; Fy	=	240	N/mm^2	
Modulus of elasticity, E	=	200000	N/mm^2	
Width Thickness ratio ; D/t	=	11.85		
Limiting width thickness ratio for compact section, $Y_p = 0.07 E/Fy$	=	58.33		(Table 2.5_1:SBC-306))
	>	11.85		
Flexural design strength of post = $\emptyset c P_n$				(Cl. No 5.2.1 of SBC-306:2007)
$\emptyset c$ = 0.85				
P_n = $A_g F_{cr}$				
A_g = gross area of member = $\Pi/4 (D^2-D1^2)$	=	431.93	mm^2	
Moment of Inertia, I = $\Pi /64(D^4-D1^4)$	=	81296.51	mm^4	
Radius of gyration about the axis of buckling , r = $\sqrt{I / A}$	=	13.72	mm	
Effective length factor , K	=	1		
Laterally unbraced length of member, l	=	1100	mm	
$\lambda_c = (Kl / r\Pi) \sqrt{Fy / E}$	=	0.88	<	1.5
λ^2_c				
F_{cr} = (0.658) Fy	=	172.98		
Flexural design strength of backstay = $\emptyset c P_n$ = 0.85x431.93x172.98	=	63507	N	
	>	51738		Hence, OK

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4. DESIGN OF POST FOUNDATION WITH STAAD PRO.

The foundation is modelled with sqare member element and the post is modelled with tubular section.Lateral springs are provided along the height of the foundation in contact with soil.Vertical and rotational springs are applied at the base level.

Charaterisitics of Pole pedestal :

Dimensions at the base :

Width : 35 (cm)

Height Real : 35 (cm)

Height Effective under compression : 13.22 (cm) (Refer Annexure - 1)

Pole pedestal Height :

Projection above ground : 0 (cm)

Embedded length : 85 (cm)

Applied loads on the pole pedestal :

Self weight of pole pedestal = 0.785x0.35x0.35x0.85x2.3 = 0.19 (Ton) (Unit wt of concrete = 2.3 T/m3)

Considering weight of pipe & chainlink for a span of 3m = 0.05 (Ton)

Vertical Load , N : 0.19+ 0.05 = 0.24 (Ton)

Horizontal Load , Q : 0.086 Ton/m

5. SOIL PARAMETERS

Co-efficient of horizontal modulus of subgrade reaction = 2090 Ton/m3 (As per table-12 of Geo-technical report)

K lateral at Z= 0.85 m 2090 Ton/m3

Lateral Area #1 = 0.0298

Lateral Area #2 = 0.0149

Vertical Position	Coordinate Z (m)	Soil K lateral (Ton/m3)	Spring	K lateral (Ton/m)
0	0.000	0		0.000
0.1	0.085	209		6.218
0.2	0.170	418		12.436
0.3	0.255	627		18.653
0.4	0.340	836		24.871
0.5	0.425	1045		31.089
0.6	0.510	1254		37.307
0.7	0.595	1463		43.524
0.8	0.680	1672		49.742
0.9	0.765	1881		55.960
1.0	0.850	2090		31.089

Co-efficient of vertical modulus of subgrade reaction = 2820 Ton/m3 (As per table-12 of Geo-technical report)

Vertical spring constant at the node at base level considering effective area = (0.35x0.1322x2820) = 130.5 Ton/m

Co-efficient of rotational modulus of subgrade reaction = 2820 x 2 = 5640 Ton/m3

(Considered to be 2 times of vertical modulus of subgrade reaction value i.e 2820 Ton/m3 as per table -12 of Geo-technical report)

Moment of Inertia of effective area under compression = $(0.35x0.1322^3)/12$ = 0.0000674 m4

Rotational spring = 5640 x 0.0000674 = 0.38 Ton-m / degree

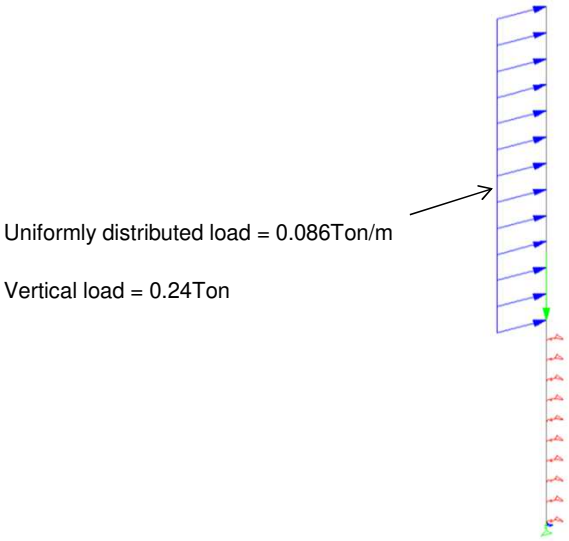
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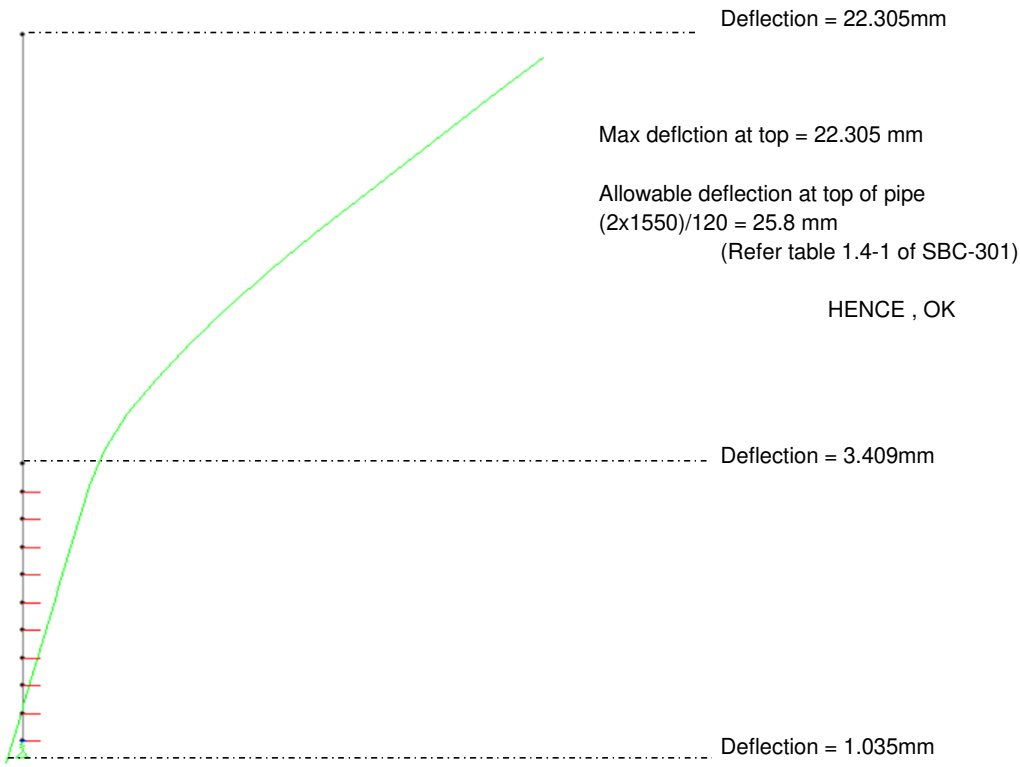
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6. DESIGN DIAGRAMS



STAAD MODEL



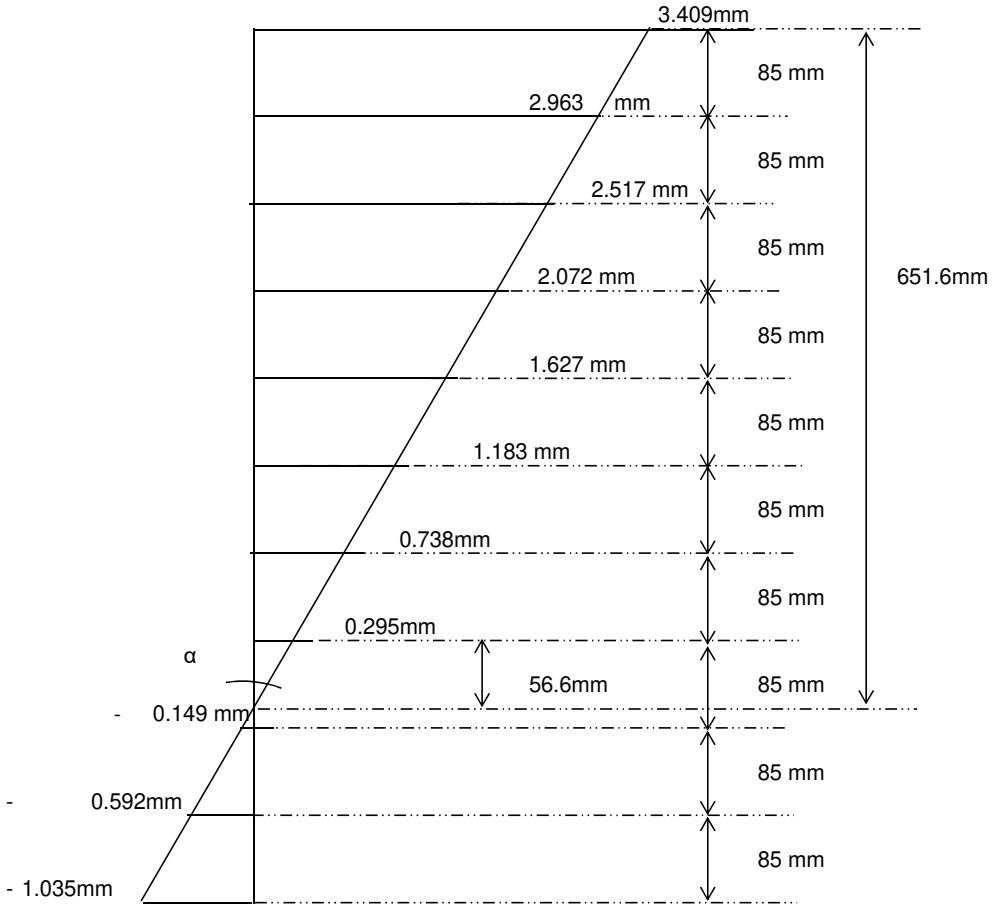
DEFLECTION DIAGRAM

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NODAL DISPLACEMENT VALUES

Tanα = 3.409 / 651.6 = 0.005 < 0.01

As per Sulzberger guidelines for slope gradient tanα <0.01, soil has elastic behaviour)

HENCE O.K.

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7. STAAD FILE

PAGE NO. 1

```
**************************************************
*                                                *
*      STAAD.Pro V8i SELECTseries5               *
*      Version  20.07.10.64                      *
*      Proprietary Program of                    *
*      Bentley Systems, Inc.                     *
*      Date=    NOV 27, 2019                     *
*      Time=    9:34:10                          *
*                                                *
*    USER ID: LNT ECC DIV                        *
**************************************************
```

```
  1. STAAD PLANE
INPUT FILE: POST FOUNDATION.STD
  2. START JOB INFORMATION
  3. ENGINEER DATE 8.11.2019
  4. JOB NAME POST FOOUNDATION
  5. ENGINEER NAME SOR
  6. CHECKER NAME SJM
  7. APPROVED NAME AG
  8. JOB REV 0
  9. JOB CLIENT MAADEN
 10. JOB NO O19045
 11. END JOB INFORMATION
 12. INPUT WIDTH 79
 13. UNIT METER MTON
 14. JOINT COORDINATES
 15. 984 0 5.87 0; 992 0 7.42 0; 1675 0 5.02 0; 1676 0 5.105 0; 1677 0 5.19 0
 16. 1678 0 5.275 0; 1679 0 5.36 0; 1680 0 5.445 0; 1681 0 5.53 0; 1682 0 5.615 0
 17. 1683 0 5.7 0; 1684 0 5.785 0
 18. MEMBER INCIDENCES
 19. 2200 984 992; 2201 1675 1676; 2203 1676 1677; 2204 1677 1678; 2205 1678 1679
 20. 2206 1679 1680; 2207 1680 1681; 2208 1681 1682; 2209 1682 1683; 2210 1683 1684
 21. 2211 1684 984
 22. ************************************************************
 23. DEFINE MATERIAL START
 24. ISOTROPIC CONCRETE
 25. E 2.21467E+006
 26. POISSON 0.17
 27. DENSITY 2.3
 28. ALPHA 1E-005
 29. DAMP 0.05
 30. TYPE CONCRETE
 31. STRENGTH FCU 2812.28
```

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32. ISOTROPIC STEEL
33. E 2.09042E+007
34. POISSON 0.3
35. DENSITY 7.83341
36. ALPHA 1.2E-005
37. DAMP 0.03
38. TYPE STEEL

STAAD PLANE -- PAGE NO. 2

39. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
40. END DEFINE MATERIAL
41. MEMBER PROPERTY AMERICAN
42. 2200 TABLE ST PIPE OD 0.0603 ID 0.05248
43. MEMBER PROPERTY AMERICAN
44. 2201 2203 TO 2211 PRIS YD 0.35 ZD 0.35
45. CONSTANTS
46. MATERIAL STEEL MEMB 2200
47. MATERIAL CONCRETE MEMB 2201 2203 TO 2211
48. **********
49. SUPPORTS
50. 1684 FIXED BUT FY FZ MX MY MZ KFX 6.218
51. 1683 FIXED BUT FY FZ MX MY MZ KFX 12.436
52. 1682 FIXED BUT FY FZ MX MY MZ KFX 18.653
53. 1681 FIXED BUT FY FZ MX MY MZ KFX 24.871
54. 1680 FIXED BUT FY FZ MX MY MZ KFX 31.089
55. 1679 FIXED BUT FY FZ MX MY MZ KFX 37.307
56. 1678 FIXED BUT FY FZ MX MY MZ KFX 43.524
57. 1677 FIXED BUT FY FZ MX MY MZ KFX 49.742
58. 1676 FIXED BUT FY FZ MX MY MZ KFX 55.96
59. 1675 FIXED BUT FZ MX MY KFX 31.089 KFY 130.5 KMZ 0.38
60. LOAD 1 LOADTYPE NONE TITLE OPERATING LOAD
61. JOINT LOAD
62. 992 FY -0.24
63. MEMBER LOAD
64. 2200 UNI GX 0.086
65. **********
66. PERFORM ANALYSIS

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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 12 NUMBER OF MEMBERS 11
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 10

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 11/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 1, TOTAL DEGREES OF FREEDOM = 36
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 188950.2 MB

67. FINISH
STAAD PLANE -- PAGE NO. 3

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 27,2019 TIME= 9:34:14 ****

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************************************************************
*  For technical assistance on STAAD.Pro, please visit     *
*  http://selectservices.bentley.com/en-US/                *
*                                                          *
*  Details about additional assistance from                *
*  Bentley and Partners can be found at program menu       *
*  Help->Technical Support                                 *
*                                                          *
*     Copyright (c) 1997-2014 Bentley Systems, Inc.        *
*             http://www.bentley.com                       *
************************************************************
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8. ANNEXURE - 1 (Calculation for effective compression height at base)

Soil Properties :

Lateral ballast coefficient at baseline; C_L : 2.09 (Kg/cm3) (As per table-12 of Geo-technical report)

Base line rotational ballast coefficient; $C_{Ø}$: 5.64 (Kg/cm3) (Considered to be 2 times of vertical modulus of subgrade reaction value i.e 2.82 Kg/cm3 as per table -12 of Geo-technical report)

Charaterisitics of Pole pedestal :

Dimensions at the base :

Width : 35 (cm)

Height Real : 35 (cm)

Height Effective under compression : 13.22 (cm)

Pole pedestal Height :

Projection above ground : 0 (cm)

Embedded length : 85 (cm)

Applied loads on the pole pedestal :

Self weight of pole pedestal = 0.785x0.35x0.35x0.85x2.3 = 0.19 (Ton) (Unit wt of concrete= 2.3 Ton/m3)

Considering weight of pipe & chainlink for a span of 3m = 0.05 (Ton)

Vertical Load , N : 0.19+ 0.05 = 0.24 (Ton)

Horizontal Load , Q : 0.13 (Ton)

Bending Moment, M : 0.103 (Ton-m) (Unfactored moment)

Ys = Upper displacement = 0.82 cm

Yi = Lower displacement = -0.35 cm

Angle of rotation of pole foundation = 0.01

Moment at stack base 0.005 (Ton-m)

Pressure at base

Trapezoidal diagram

max. Pressure = 1.027 kg/cm2

min. Pressure = 0.002 kg/cm2 > 0 Hence O.K.

Calculation area :

Determination of Displacements

Moment of Inertia	6738.78	Af =	462.7	Only compressed section
Rigidity Lateral	73.15	Wf =	1019.48	Only compressed section
Stiffness	38006.71			

Stiffness Matrix

44490	43595	Ys	21350
1	2	Yi	0.1254

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REFERENCES

[1] Structural Loading and Forces ; SBC_Code_301_2007

[2] Saudi Building Code, Steel Structures Requirements ; SBC 306-_2007

[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001

[4] Standard Specification for Pipe,Steel, Black and Hot-dipped,Zinc coated, Welded and Seamless ; ASTM designation A 53/A 53M – 99b

[5] Standard Practice for Installation of Chain-Link Fence ; ASTM F567 – 11a

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ATTACHMENT 1 : Excerpts from Geo-technical Report

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ATTACHMENT 2 : Comment Resolution Log Sheet / Compliance Report

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